



Department of Computer Science
Assignment 04

Course Code: CS-352
Course Title: Artificial Intelligence
Total Marks: 15
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Name: _____
Registration No. _____
Discipline: BS(CS)-3rd B
Submission Date: 09-02-2026

[CLO3, GA4, C4]

Question:

An online shopping platform receives thousands of customer reviews every day, making it difficult for management to manually understand customer opinions. To solve this problem, the company decides to design an automated intelligent system using Artificial Intelligence and Natural Language Processing (NLP) that can analyze customer feedback and determine overall sentiment to support business decisions.

You are provided with the following customer reviews:

1. **“I purchased this product last week and I am extremely satisfied because the quality is excellent and the overall performance exceeded my expectations.”**
2. **“Although the delivery was on time, I am disappointed with the product because the material feels cheap and the experience was very frustrating.”**

Tasks: *Apply* the following:

a) Use NLP preprocessing steps to both reviews and show the output after each step:

- Sentence tokenization
- Word tokenization
- Stop-word removal
- Stemming or lemmatization

b) Using the cleaned text, perform sentiment analysis and classify each review as positive or negative, giving a brief reason.

c) Apply TF-IDF to extract important words from the reviews and represent each review as a numerical feature vector.

Note: consider important sentiment-related terms from part a.

d) Analyze how the generated feature vectors and sentiment results help the company build an automated AI-based decision-making system.